

P4 — Evaluating the Quality of Explanations

Explainable and Trustworthy AI Course

Politecnico di Torino — 2025/2026

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Project. This project aims to explore the evaluation of explanation methods, focusing on how to assess the quality, reliability, and usefulness of explanations provided by explainable AI techniques. In addition to analyzing existing approaches, the project will involve proposing new evaluation strategies to better capture different aspects of explanation quality.

Overview.

Explanation methods are designed to provide insights into machine learning model decisions. However, evaluating the quality of these explanations remains a challenging and open problem. While many approaches exist [2, 4, 5, 7, 6, 9, 13, 12, 14], they often focus on specific aspects such as robustness or sensitivity, without providing a comprehensive view of explanation quality. Explanations should be assessed across multiple dimensions, including faithfulness to the model, plausibility for humans, robustness to perturbations, and usability in real-world contexts. A systematic evaluation is therefore needed to move from anecdotal evidence toward quantitative and comparable assessment of explanations.

Different evaluation criteria capture complementary aspects of explanation quality. For instance, faithfulness assesses whether the explanation reflects the true behavior of the model, while plausibility evaluates whether it aligns with human reasoning. Other properties, such as consistency, continuity, and compactness, further contribute to understanding the reliability and usefulness of explanations. Despite the variety of proposed metrics and evaluation frameworks [10, 3, 1, 8, 11], existing tools often cover only a subset of these aspects, and there is still no unified approach to evaluate explanations in a comprehensive and systematic way. This project aims not only to analyze these limitations but also to explore novel evaluation strategies that better capture the multi-faceted nature of explanation quality.

Goal.

The goal of this project is to study and systematize the evaluation of explanation methods by analyzing multiple dimensions of explanation quality. The project aims to identify limitations of current evaluation strategies and to propose new or improved evaluation methods that provide a more comprehensive and meaningful assessment of explanations.

Required analysis, implementation, and evaluation.

- **Literature Review.** Conduct a systematic review of existing evaluation methods for explanation quality. The review should cover different dimensions such as faithfulness, plausibility, robustness, stability, and compactness, as well as different evaluation paradigms, including perturbation-based methods, sanity checks, synthetic data evaluations, and user studies. The review should analyze how these methods capture different aspects of explanation quality and where they fall short.
- **Identification of Research Gaps.** Identify key limitations in current evaluation approaches. For example, many methods focus on a single property of explanations, such as robustness or sensitivity, without considering other aspects such as usability or human interpretability. Other challenges include the lack of standardized benchmarks, the difficulty of comparing different explanation methods, and the gap between quantitative metrics and human-centered evaluation. Based on this analysis, define a clear direction for improving explanation evaluation
- **Implementation.** Select a set of explanation methods and evaluation techniques to analyze. Design and implement an experimental framework that allows evaluating explanations across multiple dimensions. As part of this phase, propose and implement at least one novel evaluation method or metric, or extend an existing one to better capture an aspect of explanation quality that is currently underexplored (e.g., combining faithfulness and plausibility, or integrating user-oriented criteria).
- **Evaluation.** Assess the behavior of explanation methods under the proposed evaluation framework. Compare existing metrics with the proposed evaluation approach. Evaluate whether the proposed evaluation framework and metric(s) provide additional insights compared to existing metrics and discuss their strengths, limitations, and applicability.

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